

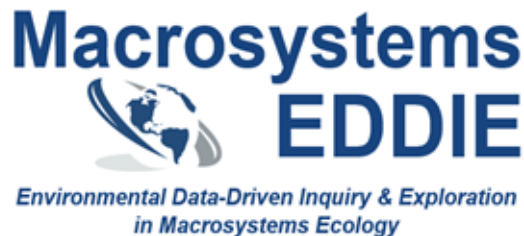
Macrosystems EDDIE: Getting Started + Troubleshooting Tips

Developed by K.J. Farrell and C.C. Carey
for use with Macrosystems EDDIE modules.

<http://module1.macrosystemseddie.org>

Module development supported by NSF EF 1702506.

Last updated: 3 Mar. 2026 by M.E. Lofton



R and RStudio



R

- Statistical environment



RStudio

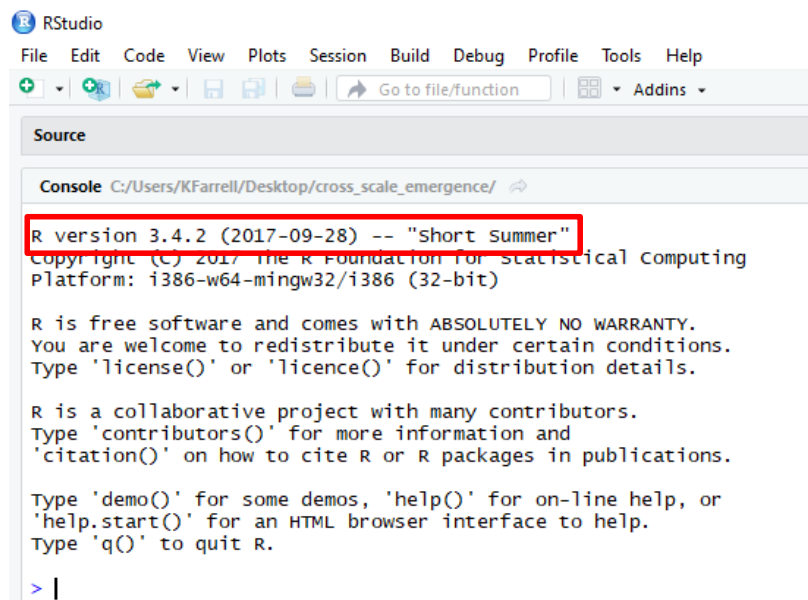
- Point and click program for using R in one place
 - Run code
 - Visualize plots
 - Access files

Check-in:

- Have you downloaded **both** R and RStudio?
- Look in your Applications (Mac) or in the Start menu (Windows) to confirm this-- both programs should be listed.
- If either program is missing, install it now!

Are R & RStudio up to date?

- Check that R and RStudio are both up-to-date, and download new versions if necessary
 - When you open RStudio, you will see your version of R. It should be at least **3.5.1**
 - Check for updates to RStudio by clicking **Help**, then **Check for Updates**



The screenshot shows the RStudio application window. The menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. The toolbar contains icons for file operations and a search bar. The Source pane is empty. The Console pane shows the output of the R startup process, with the first line, "R version 3.4.2 (2017-09-28) -- 'Short Summer'", highlighted with a red rectangle. The text in the console includes the R version, copyright information, and instructions on how to use R.

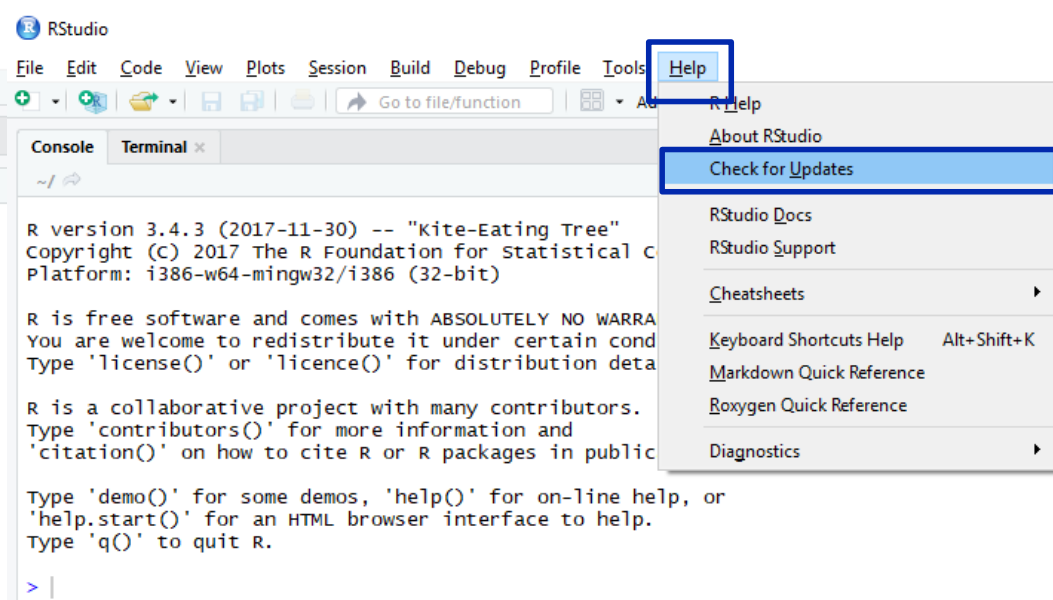
```
R version 3.4.2 (2017-09-28) -- "Short Summer"
Copyright (C) 2017 The R Foundation for Statistical Computing
Platform: i386-w64-mingw32/i386 (32-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```



The screenshot shows the RStudio application window with the Help menu open. The 'Help' menu item in the top bar is highlighted with a blue rectangle. The Help menu is open, showing options like 'About RStudio', 'Check for Updates', 'RStudio Docs', 'RStudio Support', 'Cheatsheets', 'Keyboard Shortcuts Help', 'Markdown Quick Reference', 'Roxygen Quick Reference', and 'Diagnostics'. The 'Check for Updates' option is highlighted with a blue rectangle. The console output shows the R version 3.4.3 (2017-11-30) -- 'Kite-Eating Tree'.

```
R version 3.4.3 (2017-11-30) -- "Kite-Eating Tree"
Copyright (C) 2017 The R Foundation for Statistical Computing
Platform: i386-w64-mingw32/i386 (32-bit)

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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```

Download the module files

- Navigate to the Macrosystems EDDIE Module 1 website
 - <http://module1.macrosystemseddie.org>
- Scroll down to Teaching Materials and click Files for Running the Module

Teaching Materials:

- [Files for Running the Module](#) (Zip Archive 654kB Feb6 18)– Zipped folder of all files needed to run the module in RStudio
- [R You Ready for EDDIE? Module 1](#) (Microsoft Word 2007 (.docx) 30kB Feb6 18)– Step-by-step guide to download R, RStudio, and module files

- Save the .zip folder to your Desktop

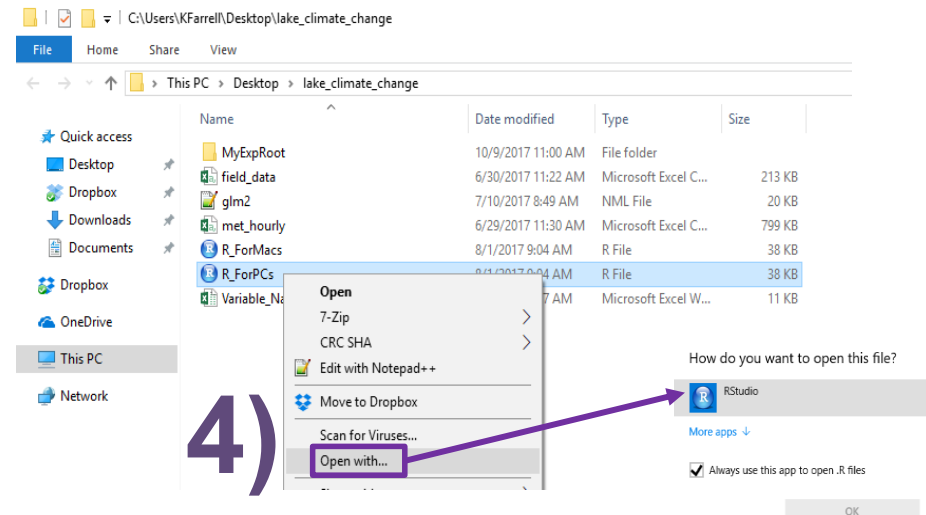
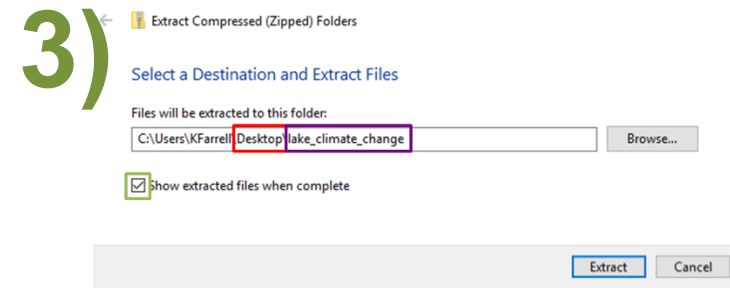
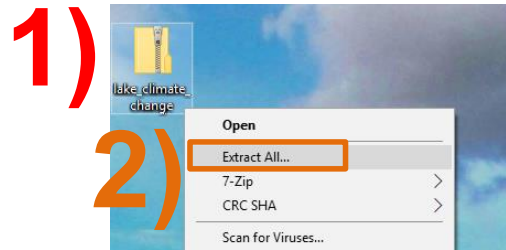
Unpack Files to Desktop: Windows

- 1) Download the zip folder directly to Desktop (or move folder from Downloads to Desktop)
- 2) Right click on the .zip folder and choose **Extract All**
- 3) Check that your files are:
 - being extracted to the **Desktop**
 - end *exactly* **lake_climate_change**.

Also **check the box**

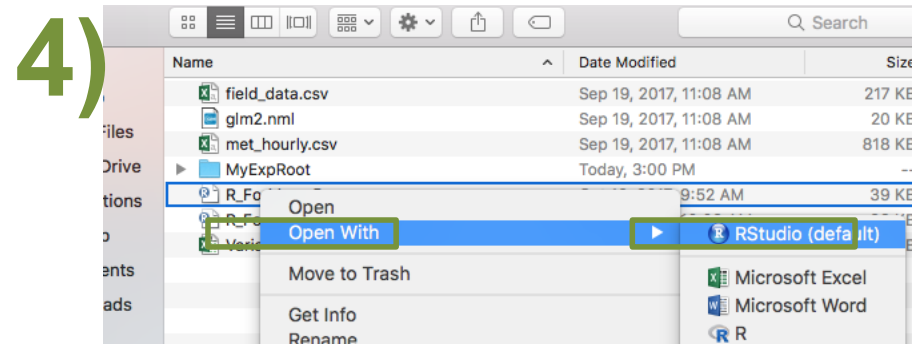
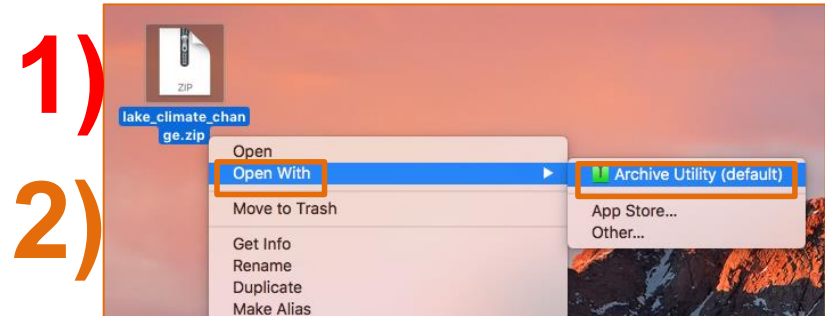
“Show extracted files when complete”

- 4) To open the module script in RStudio, right click on the file name (e.g., Module1.Rmd), then choose **Open with...** and **RStudio**



Unpack Files to Desktop: Mac

- 1) Download the zip folder directly from the Macrosystems EDDIE website to Desktop (or move folder from Downloads to Desktop)
 - **Note:** Your folder may have automatically been unzipped when you downloaded it. If it was, drag the **unzipped** 'teleconnections' folder from Downloads to the Desktop, and skip to step 4
- 2) Control + click on the .zip folder and choose **Open with** → **Archive Utility** to unzip the folder. Then double click on the unzipped folder
- 3) Check that your files are:
 - being extracted to the **Desktop**
 - called **exactly lake_climate_change**.
- 4) To open the module script in RStudio, control + click on the file name (Module1.Rmd), then choose **Open with...** and **RStudio**



Opening Module Files in RStudio

- Congrats! You've opened the module script in RStudio!

The screenshot shows the RStudio interface with the title bar 'MacrosystemsEDDIE - master - RStudio'. The 'Source' pane on the left displays the R script 'Module1.Rmd' with line numbers 582 to 591. The script contains R code for a legend and a text block about feedback. The 'Environment' pane on the right shows 'Global Environment' and 'Environment is empty'. The 'Files' pane on the bottom right shows a list of files in the 'Module 1' directory.

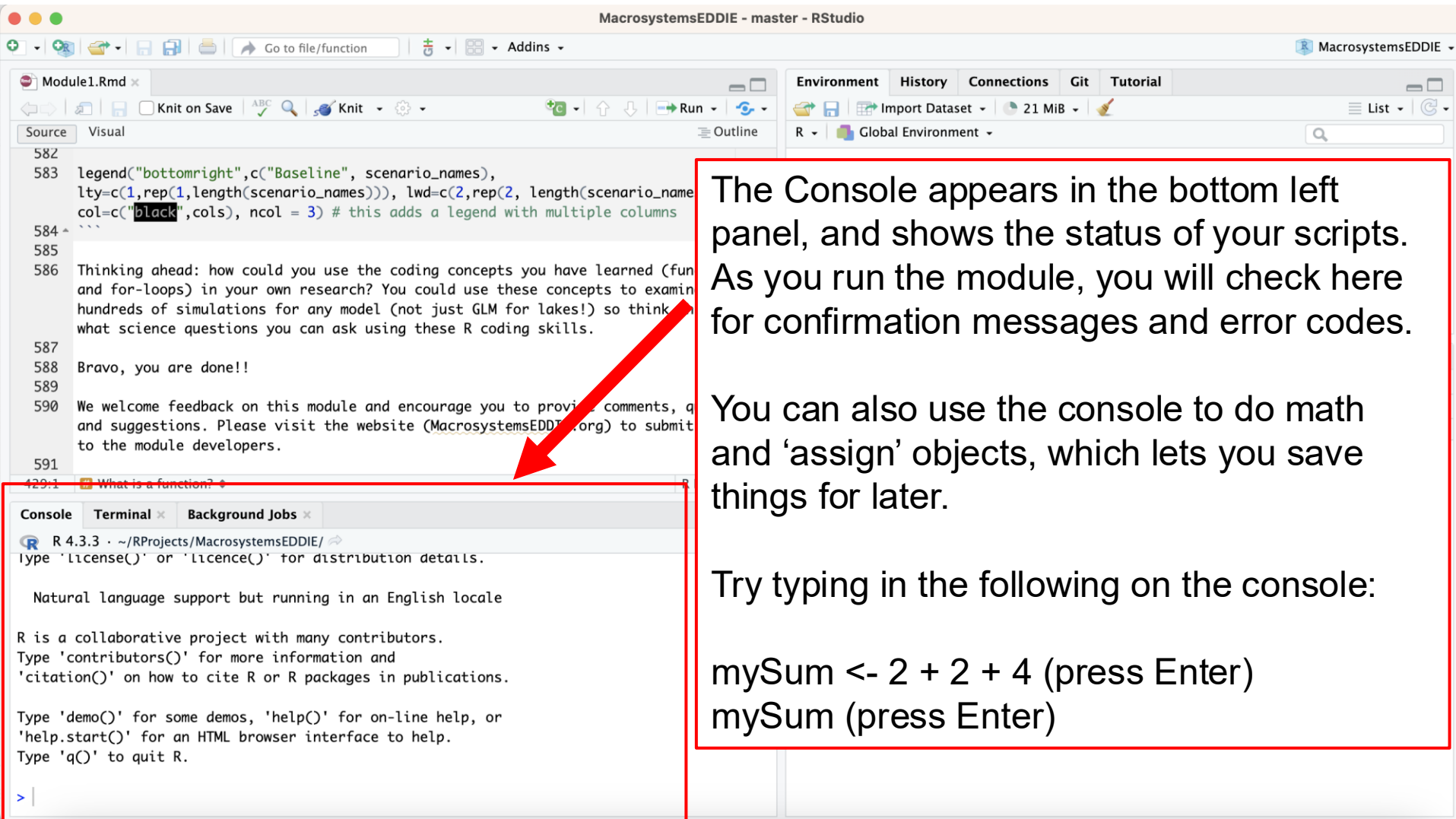
```
582  
583 legend("bottomright",c("Baseline", scenario_names),  
584       lty=c(1,rep(1,length(scenario_names))), lwd=c(2,rep(2, length(scenario_names))),  
585       col=c("black",cols), ncol = 3) # this adds a legend with multiple columns  
586  
587 Thinking ahead: how could you use the coding concepts you have learned (functions  
588 and for-loops) in your own research? You could use these concepts to examine  
589 hundreds of simulations for any model (not just GLM for lakes!) so think through  
590 what science questions you can ask using these R coding skills.  
591  
592 Bravo, you are done!!  
593  
594 We welcome feedback on this module and encourage you to provide comments, questions,  
595 and suggestions. Please visit the website (MacrosystemsEDDIE.org) to submit feedback  
596 to the module developers.  
597  
598  
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600
```

Console output:

```
R 4.3.3 ~ ~/RProjects/MacrosystemsEDDIE/  
Type 'license()' or 'licence()' for distribution details.  
  
Natural language support but running in an English locale  
  
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
>
```

Name	Size	Modified
..		
field_data.csv	206.7 KB	Dec 1, 2023, 12:54 PM
glm2.nml	2.5 KB	Oct 15, 2024, 3:56 PM
met_hourly.csv	639.5 KB	Dec 1, 2023, 12:54 PM
Module1.Rmd	36.6 KB	Oct 16, 2024, 11:30 AM
Variable_Name_Metadata.xlsx	10.3 KB	Dec 1, 2023, 12:54 PM

RStudio Basics: Console



The Console appears in the bottom left panel, and shows the status of your scripts. As you run the module, you will check here for confirmation messages and error codes.

You can also use the console to do math and 'assign' objects, which lets you save things for later.

Try typing in the following on the console:

```
mySum <- 2 + 2 + 4 (press Enter)
mySum (press Enter)
```

The screenshot shows the RStudio interface with the following details:

- Source Editor:** Contains R code for a legend and a text block. Line 582: `legend("bottomright",c("Baseline", scenario_names),`. Line 583: `lty=c(1,rep(1,length(scenario_names))), lwd=c(2,rep(2, length(scenario_name`. Line 584: `col=c("black",cols), ncol = 3) # this adds a legend with multiple columns`. Line 585: `...`. Line 586: Text block starting with "Thinking ahead: how could you use the coding concepts you have learned (fun and for-loops) in your own research? You could use these concepts to examin hundreds of simulations for any model (not just GLM for lakes!) so think what science questions you can ask using these R coding skills." Line 587: Text block starting with "Bravo, you are done!!". Line 588: Text block starting with "We welcome feedback on this module and encourage you to provide comments, q and suggestions. Please visit the website (MacrosystemsEDDIE.org) to submit to the module developers." Line 590: Text block starting with "What is a function?".
- Console:** Shows the R startup message: "R 4.3.3 · ~/RProjects/MacrosystemsEDDIE/ type 'license()' or 'licence()' for distribution details. Natural language support but running in an English locale R is a collaborative project with many contributors. Type 'contributors()' for more information and 'citation()' on how to cite R or R packages in publications. Type 'demo()' for some demos, 'help()' for on-line help, or 'help.start()' for an HTML browser interface to help. Type 'q()' to quit R. > |".

RStudio Basics: Console

The screenshot shows the RStudio interface with the 'MacroSystemsEDDIE - master - RStudio' window. The 'Source' tab is active, showing R code for a legend. The 'Environment' tab is also visible, showing the 'mySum' object with a value of 8. The 'Console' tab is at the bottom, showing the R prompt and the output of the commands entered. A red box highlights the console area, and a red arrow points to the 'mySum' object in the Environment tab.

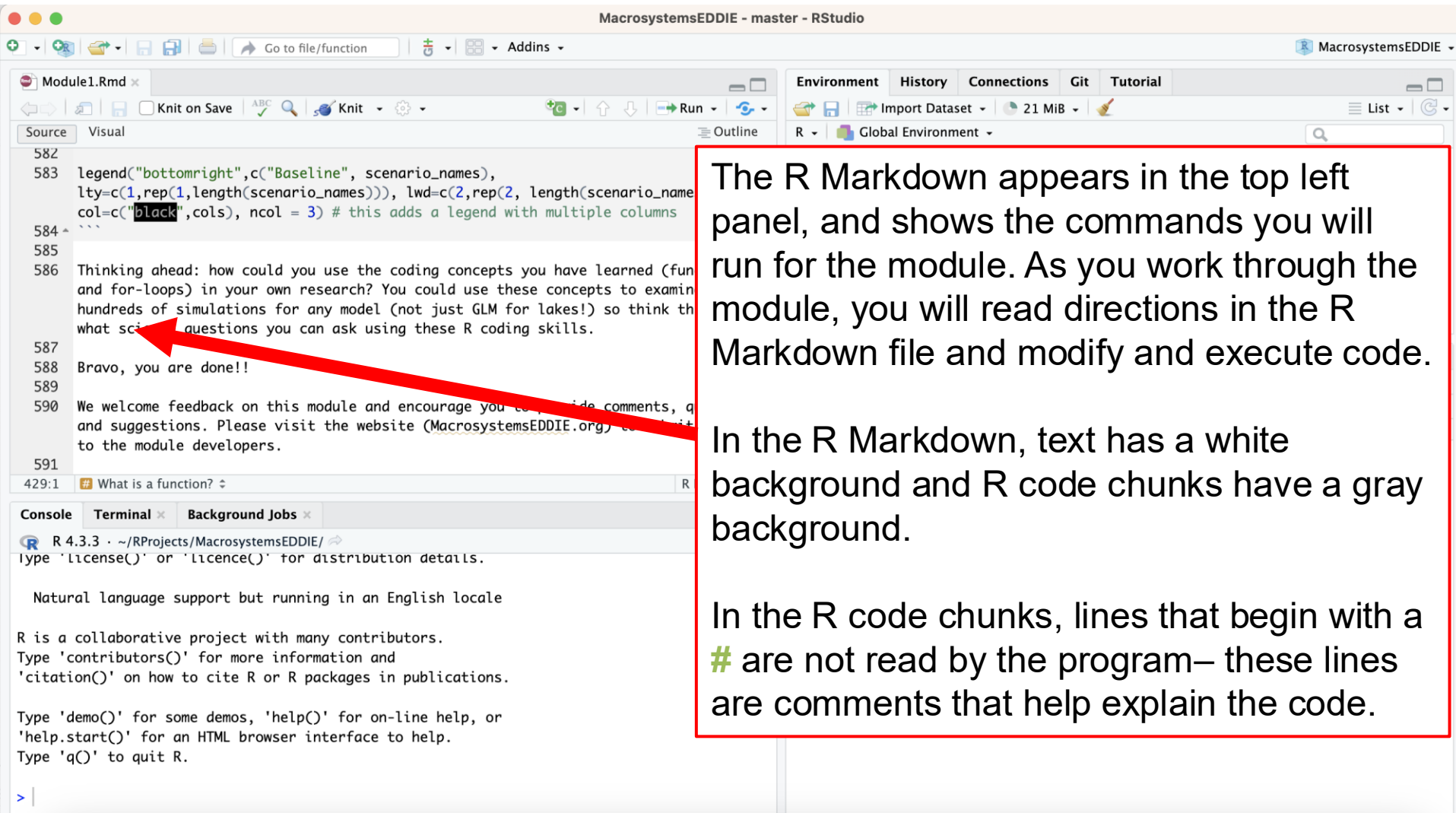
Try typing in the following on the console:

```
mySum <- 2 + 2 + 4 (press Enter)
mySum (press Enter)
```

You now see that the object mySum is saved in the Environment tab, and when you type mySum into the console, you get the answer (8)!

```
> mySum <- 2 + 2 + 4
> mySum
[1] 8
> |
```

RStudio Basics: R Markdown



The R Markdown appears in the top left panel, and shows the commands you will run for the module. As you work through the module, you will read directions in the R Markdown file and modify and execute code.

In the R Markdown, text has a white background and R code chunks have a gray background.

In the R code chunks, lines that begin with a `#` are not read by the program— these lines are comments that help explain the code.

```
582
583 legend("bottomright",c("Baseline", scenario_names),
584       lty=c(1,rep(1,length(scenario_names))), lwd=c(2,rep(2, length(scenario_name
585       col=c("black",cols), ncol = 3) # this adds a legend with multiple columns
586
587 Thinking ahead: how could you use the coding concepts you have learned (fun
588 and for-loops) in your own research? You could use these concepts to examin
589 hundreds of simulations for any model (not just GLM for lakes!) so think th
590 what sci questions you can ask using these R coding skills.
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595 and suggestions. Please visit the website (MacrosystemsEDDIE.org) to
596 to the module developers.
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Console

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R 4.3.3 · ~/RProjects/MacrosystemsEDDIE/
Type 'license()' or 'licence()' for distribution details.

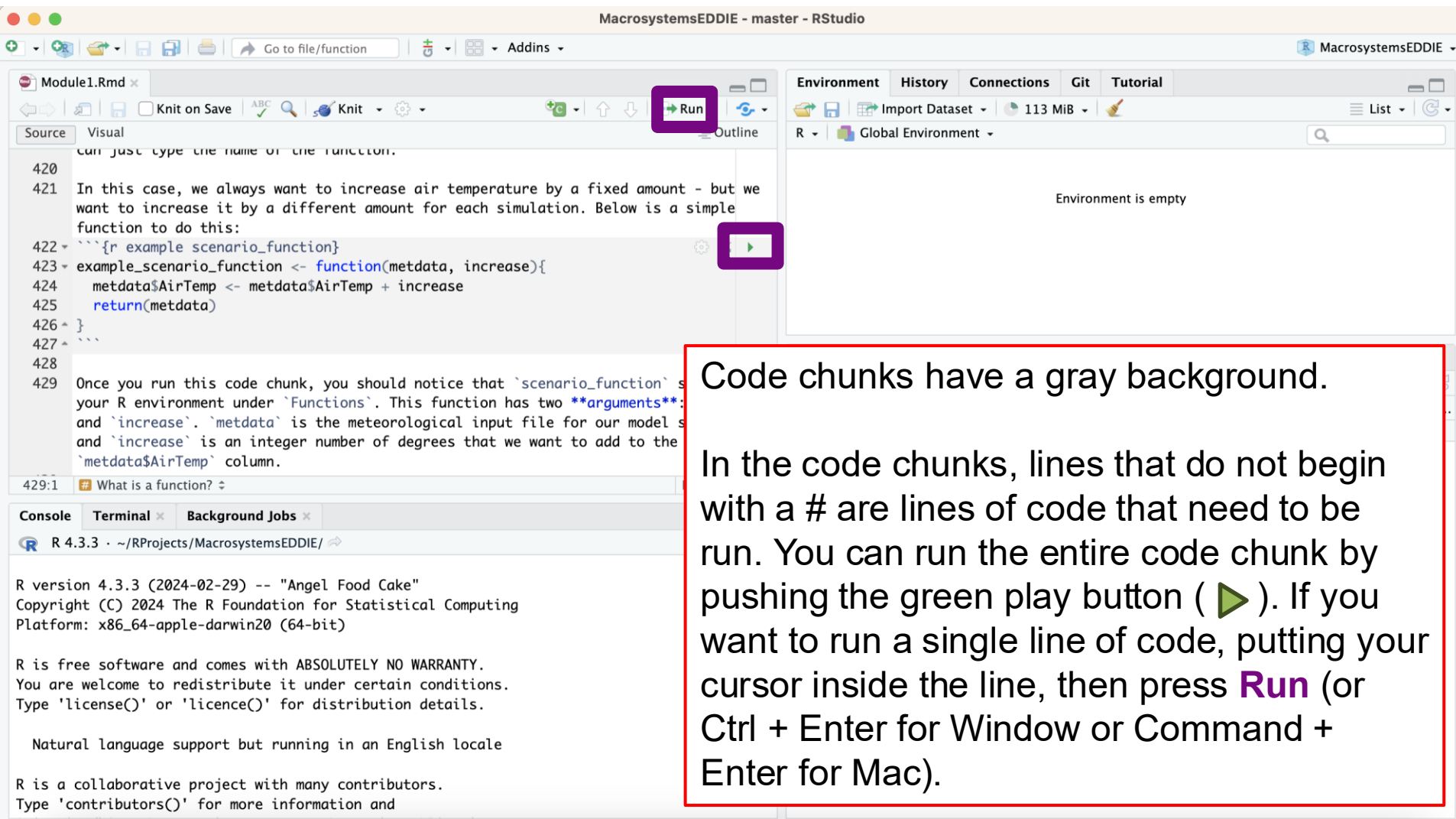
Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```

RStudio Basics: R Markdown



MacrosystemsEDDIE - master - RStudio

Module1.Rmd x

Source Visual Outline

Can just type the name of the function.

```
420
421 In this case, we always want to increase air temperature by a fixed amount - but we
    want to increase it by a different amount for each simulation. Below is a simple
    function to do this:
422 ```{r example_scenario_function}
423 example_scenario_function <- function(metdata, increase){
424   metdata$AirTemp <- metdata$AirTemp + increase
425   return(metdata)
426 }
427 ```
428
429 Once you run this code chunk, you should notice that `scenario_function` s
    your R environment under `Functions`. This function has two arguments:
    and `increase`. `metdata` is the meteorological input file for our model s
    and `increase` is an integer number of degrees that we want to add to the
    `metdata$AirTemp` column.
```

429:1 What is a function? ↕

Environment History Connections Git Tutorial

Import Dataset 113 MiB

R Global Environment

Environment is empty

Console Terminal Background Jobs

R 4.3.3 · ~/RProjects/MacrosystemsEDDIE/

R version 4.3.3 (2024-02-29) -- "Angel Food Cake"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin20 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and

Code chunks have a gray background.

In the code chunks, lines that do not begin with a # are lines of code that need to be run. You can run the entire code chunk by pushing the green play button (▶). If you want to run a single line of code, putting your cursor inside the line, then press **Run** (or Ctrl + Enter for Window or Command + Enter for Mac).

RStudio Basics: Files

The screenshot displays the RStudio IDE interface. The main editor window on the left shows a script named 'Module1.Rmd' with R code. The right-hand side of the interface contains several panels. The 'Files' panel, located at the bottom right, is highlighted with a red box and a purple bracket. It shows the file structure of the current project, 'MacrosystemsEDDIE', specifically the 'Module 1' folder. The file list includes 'field_data.csv', 'glm2.nml', 'met_hourly.csv', 'Module1.Rmd', and 'Variable_Name_Metadata.xlsx'. The 'Environment' panel above it is empty, and the 'Console' panel at the bottom left shows the R prompt.

MacrosystemsEDDIE - master - RStudio

Module1.Rmd

```
420 can just type the name of the function.
421 In this case, we always want to increase air temperature by a fixed amount - but we
422 want to increase it by a different amount for each simulation. Below is a simple
423 function to do this:
424 ```{r example scenario_function}
425 example_scenario_function <- function(metdata, increase){
426   metdata$AirTemp <- metdata$AirTemp + increase
427   return(metdata)
428 }
429 ```
```

Environment History Connections Git Tutorial

Import Dataset 113 MiB

Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

New Folder New Blank File Delete Rename More

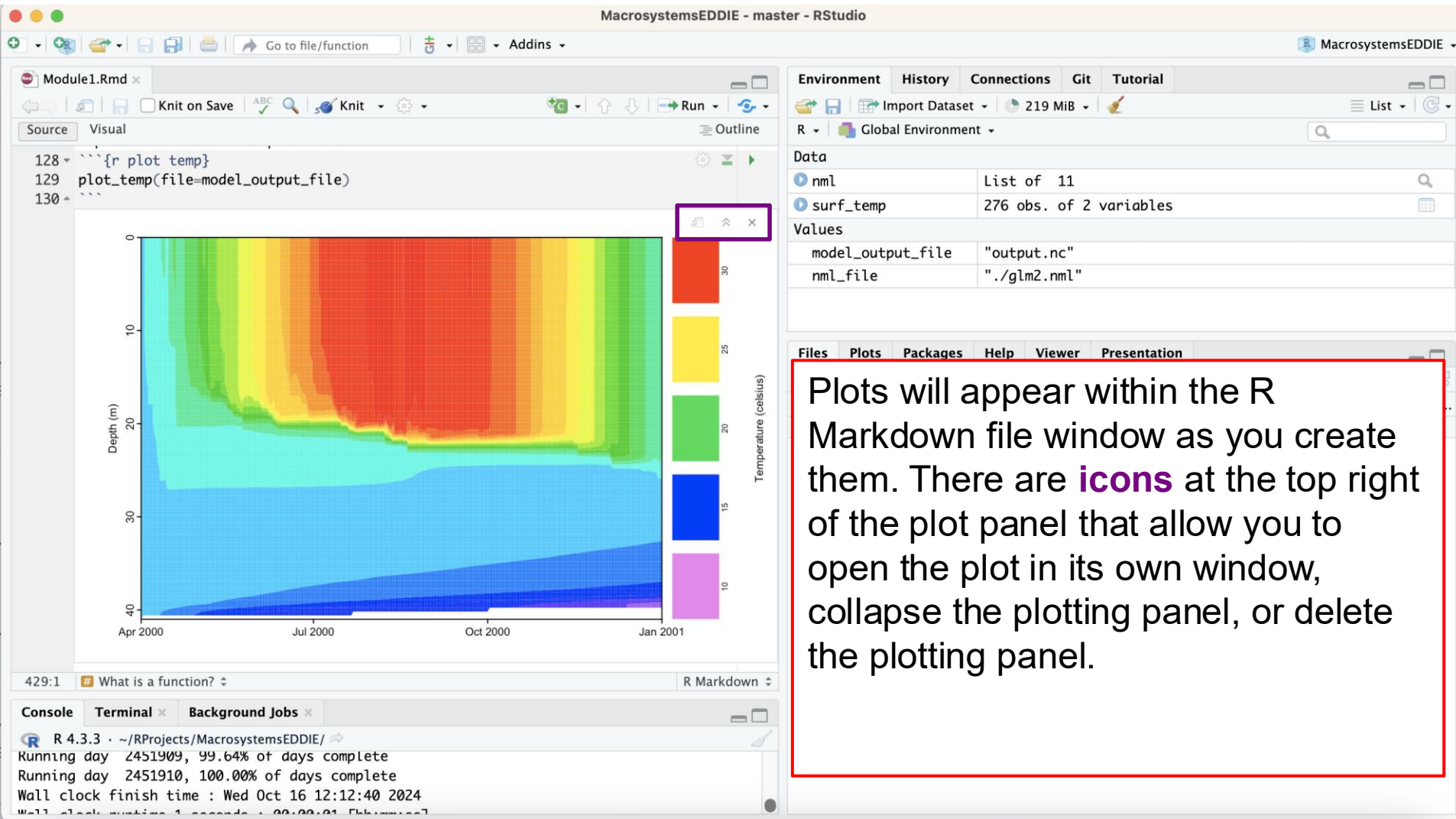
Home > RProjects > MacrosystemsEDDIE > Module 1

Name	Size	Modified
..		
field_data.csv	206.7 KB	Dec 1, 2023, 12:54 PM
glm2.nml	2.5 KB	Oct 15, 2024, 3:56 PM
met_hourly.csv	639.5 KB	Dec 1, 2023, 12:54 PM
Module1.Rmd	36.6 KB	Oct 16, 2024, 11:30 AM
Variable_Name_Metadata.xlsx	10.3 KB	Dec 1, 2023, 12:54 PM

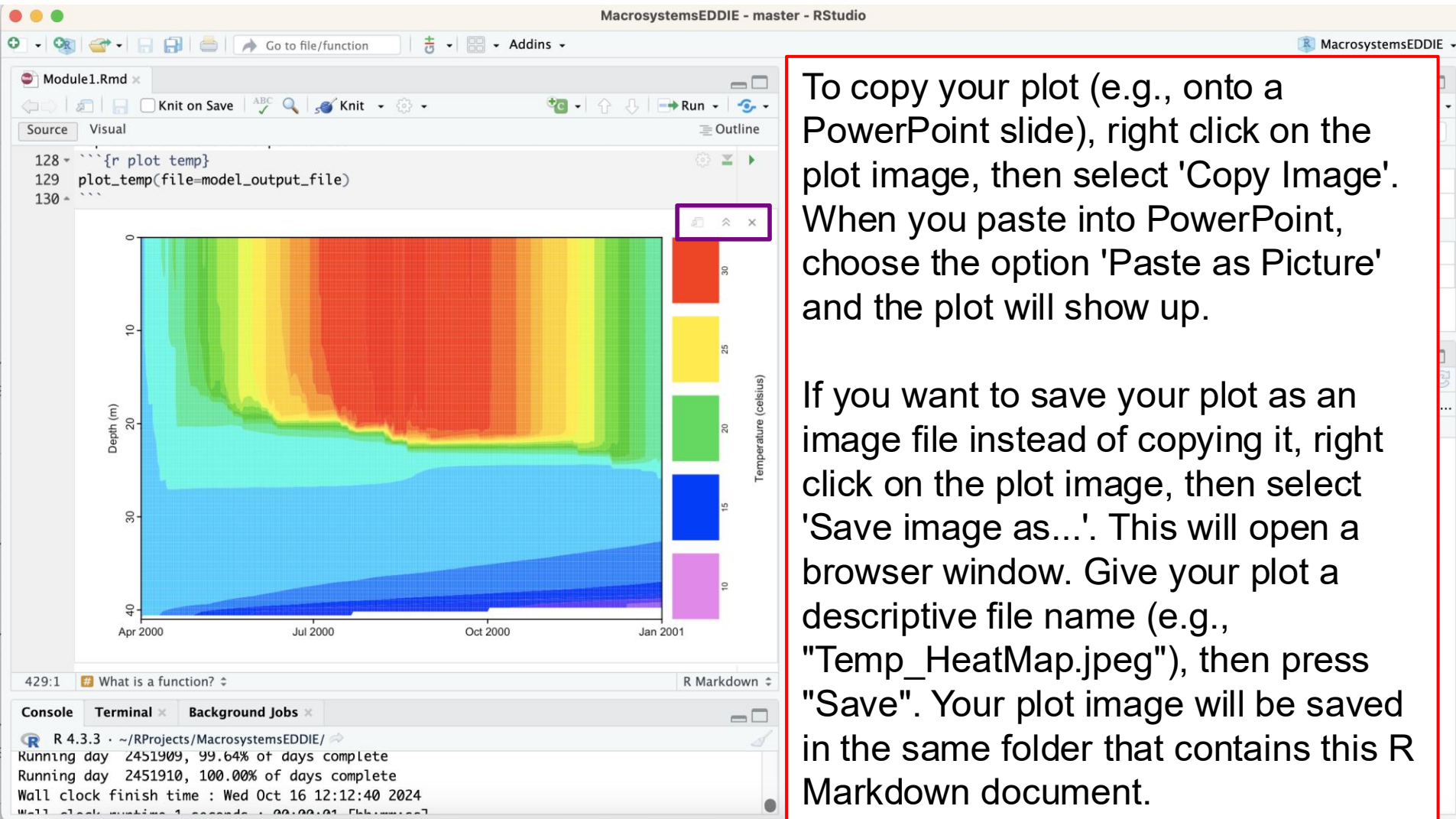
In the right panels (usually the bottom right), you will see a tab named **Files**.

This panel shows the all of the files and subfolders of your Module 1 folder.

RStudio Basics: Plots



RStudio Basics: Plots



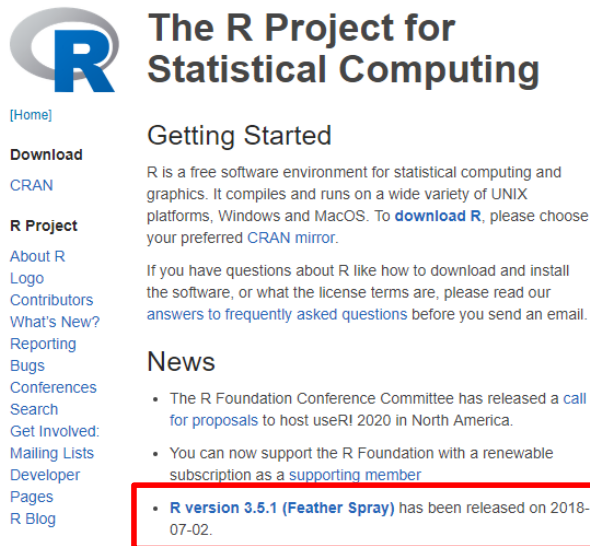
MACROSYSTEMS EDDIE: GLM TROUBLESHOOTING TIPS



Having trouble?

If you're having trouble running the Macrosystems EDDIE module, first double-check that you have the latest version of R!

- Go to <https://www.r-project.org/> and make sure that the version listed on the home page matches the version that opens when you open RStudio



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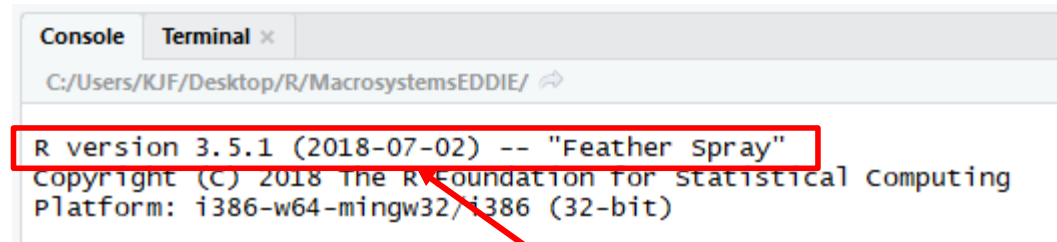
Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred [CRAN mirror](#).

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- The R Foundation Conference Committee has released a [call for proposals](#) to host useR! 2020 in North America.
- You can now support the R Foundation with a renewable subscription as a [supporting member](#)
- **R version 3.5.1 (Feather Spray)** has been released on 2018-07-02.



```
Console Terminal x
C:/Users/KJF/Desktop/R/MacrosystemsEDDIE/
R version 3.5.1 (2018-07-02) -- "Feather Spray"
Copyright (C) 2018 The R Foundation for Statistical Computing
Platform: i386-w64-mingw32/i386 (32-bit)
```

These versions **must** match!
Otherwise you'll get error messages when
downloading packages to run the GLM model

- If it doesn't match, close RStudio, download and install the new version of R, then reopen RStudio and the Teleconnections_R_Script.R file

Error: glm.exe had status 309

When does it happen?

- `run_glm(sim_folder, verbose=TRUE)` will start the GLM run, but you will likely get an error similar to: “glm.exe had status 309”

Why?

- Problem with 32-bit vs. 64-bit R in Windows 10

How to fix it:

- 1) In the RStudio menu, click on Tools, then Global Options.
- 2) In the General tab, check what R version RStudio is using (the first line at the top of the window).
- 3) If the selected version starts with [Default] [64-bit], try pressing Change and selecting the [Default] [32-bit] option. You will then need to restart RStudio and try the script again.

Error: Day 2451636 (2000-04-01) not found

When does it happen?

- `run_glm(sim_folder, verbose=TRUE)` will start the GLM run, but you will likely get an error similar to: “Day 2451636 (2000-04-01) not found”

Why?

- *time* column in .csv file not formatted correctly for GLM

How to fix it:

- 1) Open .csv file in Excel. Right click on the *time* column, then select Format.
- 2) Choose Custom, then type in **YYYY-MM-DD HH:MM:SS** *exactly*. Save and close your .csv file.
- 3) Run the following lines in R to ensure your time column is formatted for GLM (search to find in the R script, then run):
 - `metdata <- read.csv("met_hourly_climate.csv", header=TRUE)`
 - `metdata$time <- as.POSIXct(strptime(metdata$time, "%Y-%m-%d %H:%M:%S", tz="EST"))`
 - `write.csv(metdata, "met_hourly_climate.csv", row.names=FALSE, quote=FALSE)`Make sure you edit the file name (in blue, above) to match your .csv file.

Error: "MSVCR100.dll is missing"

When does it happen?

- When you try to run GLM commands, you receive the error: "MSVCR100.dll is missing from your computer" or "The code execution cannot proceed because MSVCR100.dll was not found. Reinstalling the program may fix this problem"

Why?

- The MSVCR100.dll file is missing from your Windows C++ library

How to fix it:

The missing library (MSVCR100.dll) will need to be reinstalled on your computer. This is beyond the scope of Macrosystems EDDIE troubleshooting, and we recommend you check with a campus IT worker for help.

In the meantime, we recommend partnering with a student whose computer isn't having this problem to run the Macrosystems EDDIE module.